

The Dot Product

One number that says how much two vectors agree · video 1:52

This lesson has one big idea: the **dot product** turns two vectors into **one number** that says how much they point the same way. The recipe is multiply matching numbers, then add. Divide by both lengths and you get **cosine similarity**, a pure measure of direction. LLMs use dot products almost everywhere.

How to use this guide: master one concept before moving to the next.

1. Watch the whole video once, without stopping.
2. Then take the concepts **in order**. For each one: read the short explanation, look at the picture from the video (▶ is the time in the video), and answer every question in its *Check* without looking anything up.
3. Check your answers in the **Answers** section at the end (page 9).
4. **Pass mark: every question in the check correct.** If you missed one, rewatch that concept's video segment, reread the explanation, and redo only the questions you missed. Tick the concept off below, then move on.

It is strongly suggested that you **master each concept before starting the next one**. Each concept builds on the previous ones.

Concepts in this lesson

CONCEPT	VIDEO SEGMENT	QUESTIONS	MASTERED
1. Multiply matching numbers, then add	▶ 0:08-0:37	4	☐
2. Positive, zero, negative: agreement	▶ 0:37-0:50	4	☐
3. The shadow and the angle	▶ 0:50-1:07	4	☐
4. Cosine similarity	▶ 1:07-1:21	4	☐
5. Dot products everywhere	▶ 1:21-1:33	4	☐
6. The dot product in NumPy	▶ 1:33-1:49	3	☐

Before you start: this builds on F01 (Vectors): lists of numbers, arrows, and length by Pythagoras. The dot product is used in main episodes 3 (Embeddings), 5 and 6 (Attention), 8 (The MLP) and 10 (From Vectors Back to Words). Concept 6 reads a few lines of NumPy.

Multiply matching numbers, then add

Given two vectors, a useful question is: **how much do they point the same way?** The **dot product**, written $\mathbf{a} \cdot \mathbf{b}$, answers it with a single number. The recipe: **multiply matching numbers, then add everything up**. For $\mathbf{a} = (3, 1)$ and $\mathbf{b} = (2, 2)$: $3 \times 2 + 1 \times 2 = 6 + 2 = 8$.

multiply matching entries, then add them up

$$\begin{array}{ccc}
 \begin{array}{c} \text{a} \\ \left[\begin{array}{c} 3 \\ 1 \end{array} \right] \end{array} & \begin{array}{c} \text{b} \\ \left[\begin{array}{c} 2 \\ 2 \end{array} \right] \end{array} & \\
 \begin{array}{c} \text{---} \longrightarrow 3 \cdot 2 = 6 \\ \text{---} \longrightarrow 1 \cdot 2 = 2 \end{array} & \cdot & \\
 \hline
 6 + 2 = \boxed{8} & &
 \end{array}$$

same recipe, any number of dimensions

a 1 768

b

same length

768 multiplications

$\Sigma \rightarrow$ one sum

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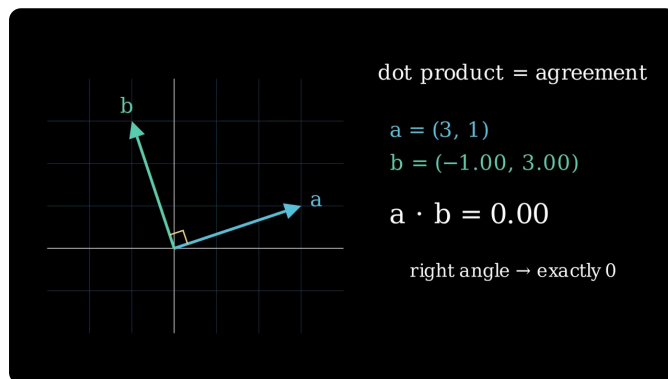
CONCEPT 2 OF 6

Positive, zero, negative: agreement

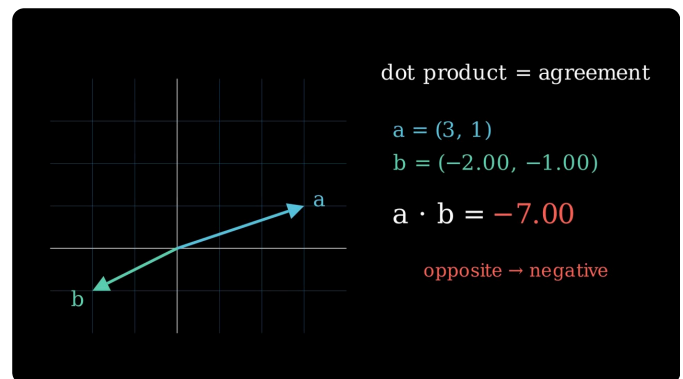
▶ Video 0:37-0:50

Geometrically, the dot product measures **agreement**. Keep $a = (3, 1)$ and turn b :

B	$A \cdot B$	THE ARROWS...
$(3, 2)$	$3 \cdot 3 + 1 \cdot 2 = \mathbf{11}$	point the same way: large positive
$(-1, 3)$	$3 \cdot (-1) + 1 \cdot 3 = \mathbf{0}$	are at right angles: exactly zero
$(-2, -1)$	$3 \cdot (-2) + 1 \cdot (-1) = \mathbf{-7}$	point in opposite directions: negative



▶ 0:46 b turned to $(-1, 3)$: at right angles to a , and $a \cdot b = 0$.



▶ 0:49 $b = (-2, -1)$ points roughly opposite to a : $a \cdot b = -7$.

KEY IDEA

$a \cdot b > 0$: they agree. $a \cdot b = 0$: at right angles. $a \cdot b < 0$: they point in opposite directions.

Check 2: Positive, zero, negative: agreement

Answer all 4 questions before looking at the answers. Pass mark: all correct.

2.1 CALCULATION

Compute $(2, 1) \cdot (-1, 2)$. What does the result tell you about the two arrows?

2.2 MULTIPLE CHOICE

$a = (1, 1)$. Which b gives the most negative $a \cdot b$?

- A. $(2, 2)$
- B. $(1, -1)$
- C. $(-2, -1)$
- D. $(-1, 0)$

2.3 TRUE OR FALSE

"If $a \cdot b$ is negative, the angle between a and b is more than 90° ."

True ☐ False ☐

2.4 CALCULATION

Find the number k that makes $(4, 2) \cdot (1, k) = 0$.

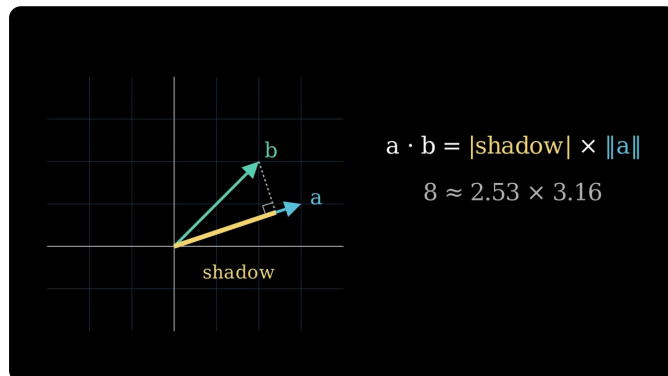
CONCEPT 3 OF 6

The shadow and the angle

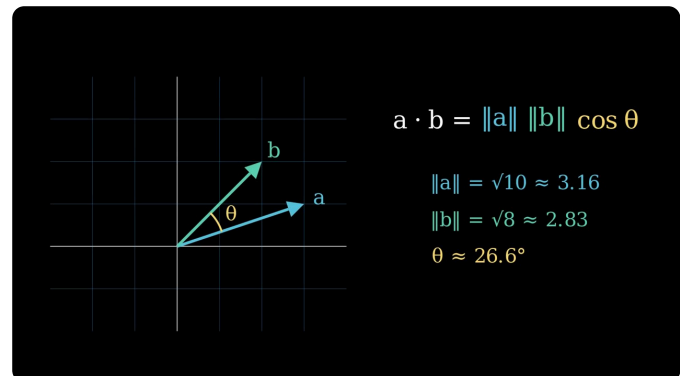
▶ Video 0:50–1:07

Here's the picture: **project** b onto a , like a shadow. The dot product is the **length of that shadow times the length of a** . For our example the shadow is about 2.53 long, a has length $\sqrt{10} \approx 3.16$, and $2.53 \times 3.16 \approx 8$.

That gives a second formula: $\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta$, where $\|\mathbf{a}\|$ is the length of a (F01) and θ is the angle between the arrows. The cosine is 1 at 0° , 0 at 90° and -1 at 180° , which is why right angles give exactly zero. Check: $3.16 \times 2.83 \times \cos 26.6^\circ \approx 3.16 \times 2.83 \times 0.894 \approx 8$.



▶ 0:59 Project b onto a like a shadow: $\mathbf{a} \cdot \mathbf{b} = \text{shadow length} \times \text{length of } a$, $8 \approx 2.53 \times 3.16$.



▶ 1:06 The second formula: $\|\mathbf{a}\| = \sqrt{10} \approx 3.16$, $\|\mathbf{b}\| = \sqrt{8} \approx 2.83$, $\theta \approx 26.6^\circ$.

KEY IDEA

$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta$: length $\times$ length $\times$ how well the directions line up. Same number as multiply-and-add, seen as geometry.

Check 3: The shadow and the angle

Answer all 4 questions before looking at the answers. Pass mark: all correct.

3.1 CALCULATION

$\|\mathbf{a}\| = 2$, $\|\mathbf{b}\| = 5$ and the angle between them is 60° ($\cos 60^\circ = 0.5$). What is $\mathbf{a} \cdot \mathbf{b}$?

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3.2 CALCULATION

$\mathbf{a} = (4, 0)$ and $\mathbf{b} = (3, 5)$. How long is b 's shadow on a ? Check that shadow $\times \|\mathbf{a}\|$ equals $\mathbf{a} \cdot \mathbf{b}$.

.....

3.3 MULTIPLE CHOICE

Two vectors both have length 3. When is their dot product largest?

- A.** When they are at right angles
- B.** When they point the same way
- C.** When they point in opposite directions
- D.** Never: it is always 9

3.4 TRUE OR FALSE

"Making a twice as long, without changing its direction, doubles $\mathbf{a} \cdot \mathbf{b}$."

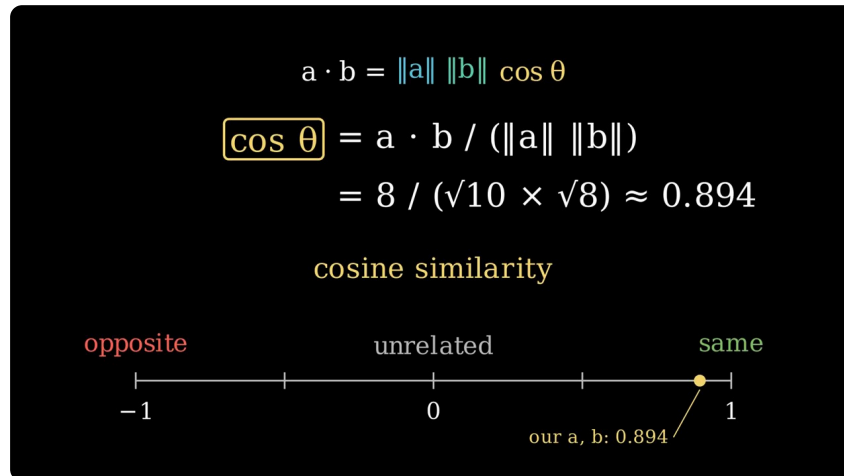
True ☐ False ☐

CONCEPT 4 OF 6

Cosine similarity

▶ Video 1:07-1:21

If we only care about **direction**, we divide the dot product by both lengths. What's left is the cosine of the angle, called **cosine similarity**: $\cos \theta = a \cdot b / (\|a\| \|b\|)$. For $a = (3, 1)$ and $b = (2, 2)$: $8 / (\sqrt{10} \times \sqrt{8}) \approx \mathbf{0.894}$. It always lies between -1 and 1 : **1** means the same direction, **0** unrelated (right angles), and **-1** opposite.



▶ 1:20 For our a and b, cosine similarity is 0.894: close to “same”.

KEY IDEA

Cosine similarity $= a \cdot b / (\|a\| \|b\|)$: the dot product with the lengths divided out. A score from **-1 to 1** for direction only.

Check 4: Cosine similarity

Answer all 4 questions before looking at the answers. Pass mark: all correct.

4.1 CALCULATION

What is the cosine similarity of $(1, 0)$ and $(1, 1)$? Round to 3 decimal places.

4.2 MULTIPLE CHOICE

Which pair has cosine similarity exactly -1 ?

- A. $(1, 2)$ and $(2, 4)$
- B. $(1, 2)$ and $(-2, 1)$
- C. $(1, 2)$ and $(-3, -6)$
- D. $(1, 2)$ and $(-1, 2)$

4.3 TRUE OR FALSE

“(3, 1) and (30, 10) have a larger cosine similarity than (3, 1) and (3, 1), because the numbers are bigger.”

True ☐ False ☐

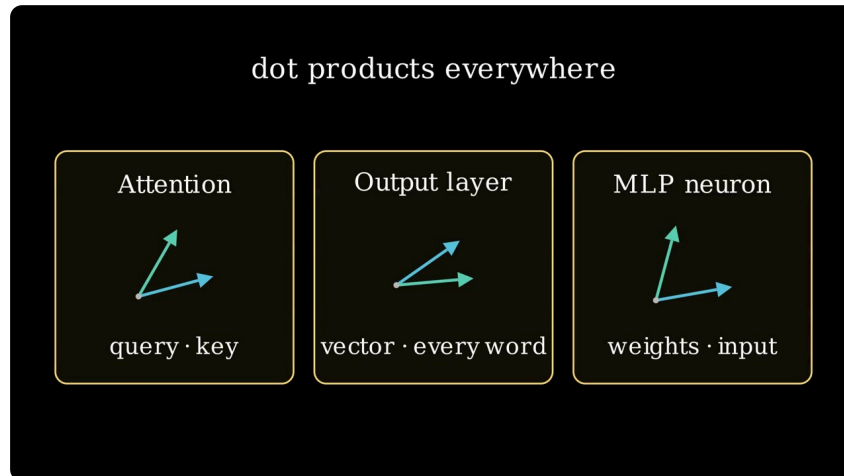
4.4 CALCULATION

What is the cosine similarity of $(3, 4)$ and $(4, 3)$?

Dot products everywhere

▶ Video 1:21-1:33

Language models use dot products everywhere. **Attention** compares a *query* vector with a *key* vector (episodes 5 and 6). The **output layer** compares the final vector with a vector for **every word** (episode 10). And each **MLP neuron** is a dot product of its weights with the input (episode 8). Each time the question is: **how much do these two vectors agree?**



▶ 1:32 Three places LLMs use dot products.

KEY IDEA

Whenever an LLM compares two vectors (query and key, vector and word, weights and input), it computes a **dot product**.

Check 5: Dot products everywhere

Answer all 4 questions before looking at the answers. Pass mark: all correct.

5.1 MULTIPLE CHOICE

In attention, a query is compared with a key. What does a large positive query · key mean?

- A. The two tokens are far apart in the text
- B. The query and key point roughly the same way: a good match
- C. The key vector is longer than the query vector
- D. The two tokens are the same word

5.2 CALCULATION

A tiny output layer knows 3 words: *cat* = (1, 2), *dog* = (2, 1), *mat* = (-1, 1). The final vector is $h = (2, 3)$. Compute $h \cdot$ each word. Which word scores highest?

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5.3 CALCULATION

An MLP neuron has weights (0.5, -1, 2) and receives the input (4, 1, 0.5). What is its dot product?

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5.4 CALCULATION

An output layer compares a 768-number vector with every word of a 50,000-word vocabulary. How many dot products is that, and how many multiplications in total?

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CONCEPT 6 OF 6

The dot product in NumPy

▶ Video 1:33-1:49

In NumPy, the dot product is the @ sign: `a @ b`. Divide by the two lengths (`np.linalg.norm`, from F01) and you get cosine similarity.

```
1 a = np.array([3, 1])
2 b = np.array([2, 2])
3 a @ b # 8
4 cos = a @ b / (np.linalg.norm(a) * np.linalg.norm(b))
5 cos # 0.894
```

▶ 1:39 The code from the video: @ gives the dot product; dividing by both lengths gives cosine similarity.

```
a = np.array([3, 1])
b = np.array([2, 2])
a @ b # 8
cos = a @ b / (np.linalg.norm(a) * np.linalg.norm(b))
cos # 0.894
```

You'll meet the dot product in episodes 3, 5, 6, 8 and 10. Next, F03 does lots of dot products at once, with matrices.

KEY IDEA

`a @ b` is the dot product; `a @ b / (norm(a) * norm(b))` is cosine similarity.

Check 6: The dot product in NumPy

Answer all 3 questions before looking at the answers. Pass mark: all correct.

6.1 MULTIPLE CHOICE

What does `a * b` (a star, not @) give for the a and b above?

- A. 8
- B. `array([6, 2])`
- C. 0.894
- D. An error

6.2 SHORT ANSWER

What does `np.array([1, 2, 3]) @ np.array([4, 5, 6])` return?

6.3 HANDS-ON CODE

Try it yourself. Below are toy 3-number “word vectors”. (a) Complete `cos_sim(a, b)`. (b) Before running: which word should come out more similar to *cat*? Then run it: what are the two numbers? (c) Replace `words["cat"]` by `10 * words["cat"]`. Which changes: the dot product, the cosine similarity, or both?

```
import numpy as np

words = {
    "cat": np.array([0.9, 0.8, 0.1]),
    "dog": np.array([0.8, 0.9, 0.2]),
    "car": np.array([0.1, 0.2, 0.9]),
}

def cos_sim(a, b):
    ...                # (a) your code here

for w in ["dog", "car"]:
    print(w, round(cos_sim(words["cat"], words[w]), 3))
```

ANSWERS

Answers and explanations

Mark each answer right or wrong. If a concept has any wrong answer, rewatch its segment (▶) and redo the questions you missed before moving on.

1. Multiply matching numbers, then add · rewatch ▶ 0:08-0:37

1.1 11. $1 \times 3 + 4 \times 2 = 3 + 8 = 11$.

1.2 (a) 6 (b) 768 multiplications, 767 additions. (a) $8 - 5 + 3 = 6$. (b) One product per position; adding up 768 numbers takes 767 “+” signs.

1.3 B. 3 numbers against 2: the third has no partner. C is fine (it is 0), and so is D (it is 3).

1.4 True. Each product is the same either way round ($3 \times 2 = 2 \times 3$), so the sum is the same.

2. Positive, zero, negative: agreement · rewatch ▶ 0:37-0:50

2.1 0: they are at right angles. $-2 + 2 = 0$, and a dot product of exactly zero means the arrows are at right angles.

2.2 C. The dot products are 4, 0, -3 and -1 . $(-2, -1)$ points most nearly opposite to a, and it is long.

2.3 True. Zero is exactly 90° ; below zero, the arrows lean away from each other.

2.4 $k = -2$. $4 \times 1 + 2 \times k = 0$, so $k = -2$: the arrow $(1, -2)$ is at right angles to $(4, 2)$.

3. The shadow and the angle · rewatch ▶ 0:50-1:07

3.1 5. $2 \times 5 \times 0.5 = 5$.

3.2 3; $3 \times 4 = 12 = a \cdot b$. a lies along the x-axis, so b's shadow is just b's x-part, 3. And $a \cdot b = 4 \times 3 + 0 \times 5 = 12$.

3.3 B. $\cos \theta$ is largest (1) at $\theta = 0^\circ$, giving $3 \times 3 \times 1 = 9$. At right angles it is 0; opposite, -9 .

3.4 True. $\|a\|$ doubles and nothing else changes. Check: $(6, 2) \cdot (2, 2) = 16$, twice 8.

4. Cosine similarity · rewatch ▶ 1:07-1:21

4.1 0.707. $a \cdot b = 1$, $\|a\| = 1$, $\|b\| = \sqrt{2}$, so $1 / \sqrt{2} \approx 0.707$ (the angle is 45°).

4.2 C. $(-3, -6) = -3 \times (1, 2)$: exactly opposite. A is +1 (same direction), B is 0 (right angles), D is 0.6.

4.3 False. Both are exactly 1: same direction. Only the raw dot product grows (100 versus 10).

4.4 0.96. $a \cdot b = 12 + 12 = 24$, and both lengths are 5, so $24 / 25 = 0.96$: very similar directions.

5. Dot products everywhere · rewatch ▶ 1:21-1:33

5.1 B. A large positive dot product means strong agreement. It says nothing about distance in the text, and length alone doesn't decide it.

5.2 cat 8, dog 7, mat 1: cat. $2 + 6 = 8$, $4 + 3 = 7$, $-2 + 3 = 1$. h points most nearly the same way as cat.

5.3 2. $0.5 \times 4 - 1 \times 1 + 2 \times 0.5 = 2 - 1 + 1 = 2$.

5.4 50,000 dot products; 38,400,000 multiplications. One dot product per word, each with 768 multiplications: $50,000 \times 768$. F03 shows how matrices do them all at once.

6. The dot product in NumPy · rewatch ► 1:33-1:49

6.1 B. `*` multiplies matching numbers but doesn't add them up. `np.sum(a * b)` is 8, the same as `a @ b`.

6.2 32. $4 + 10 + 18 = 32$.

6.3 (b) dog 0.99, car 0.303 · (c) only the dot product (a) `def cos_sim(a, b): return a @ b / (np.linalg.norm(a) * np.linalg.norm(b))`. (b) cat and dog point almost the same way (0.99); car points elsewhere (0.303). (c) The dot product with dog grows 10 times ($1.46 \rightarrow 14.6$), but the cosine similarity stays 0.99: dividing by the lengths removes the scaling.